School of Engineering,
Cochin University of Science andTechnology
B. Tech. Degree IInd Semester Second Internal Examination July 2024

23-200-0203A ENGINEERING MECHANICS (CE A Batch)
(2023 scheme)

Duration: 2 hours

Maximum Marks: 30

Course Outcomes: On completion of the course, a student will be able to

1. Explain principles and theorems related to rigid body mechanics.
2. Identify the components of a system of forces acting on the rigid body.
3. Apply the conditions of equilibrium to various practical problems involving different force system.
4. Choose appropriate theorems, principles or formulae to solve problems of mechanics.
5. Solve problems involving rigid bodies, applying the properties of distributed areas and masses.

Part A
(Answer ALL questions)

(5 x 2= 10 marks)

Q.No		QUESTIONS	Marks	BL	CO	PO
I.	(a)	Differentiate statically determinate and indeterminate beams	(2)	L1	[CO3]	1,2
	(b)	State theorems of moment of inertia	(2)	L4	[CO5]	1,2
	(c)	A car moving on a straight level road, skidded for a total distance of 60m after the brakes were applied. Determine the speed of the car just before the brakes were applied, if the coefficient of friction between the car tyres and the road is 0.4. Take g= 9.80m/s^2	(2)	L3	[CO4]	1,2
	(d)	A mortar having muzzle velocity v_0= 215.6m/s fires for maximum range across a level plain. Neglecting air resistance, calculate the time of flight of the shell	(2)	L3	[CO4]	1,2
	(e)	A wooden block weighing 22.2N rests on a smooth horizontal surface. A revolver bullet weighing 0.14N is shot horizontally into the side of the block. If the block attains a velocity of 3.05m/s, what was the muzzle velocity of the bullet.	(2)	L1	[CO4]	1,2

Part B
(Answer any TWO questions)

(2 x 10= 20 marks)

II. (a) A roller weighing 2000N rests on an inclined bar weighing 800N as shown in fig.1. Assuming weight of the bar AB is negligible, determine the reactions developed at supports C and D. (5) L5 [CO3] 1,2

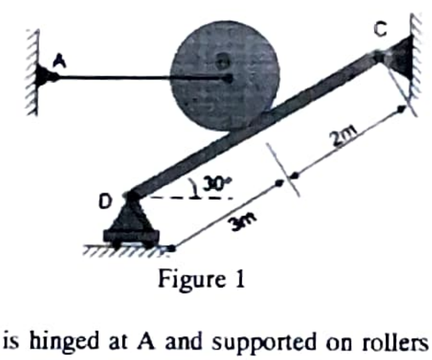

Figure 1

(b) A beam 8m long is hinged at A and supported on rollers over a smooth surface inclined at 30° to the horizontal at B. The beam is loaded as shown in fig.2. Determine the support reactions. (5) L5 [CO4] 1,2

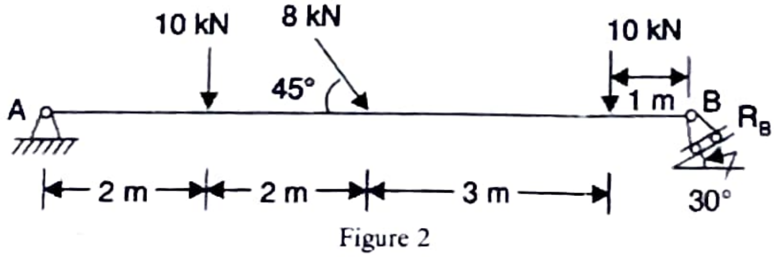

Figure 2

OR

III An I-section is made up of three rectangles as shown in fig.3. Find the moment of inertia of the section about the horizontal axis passing through the centroid of the section. (10) L5 [CO5] 1,2

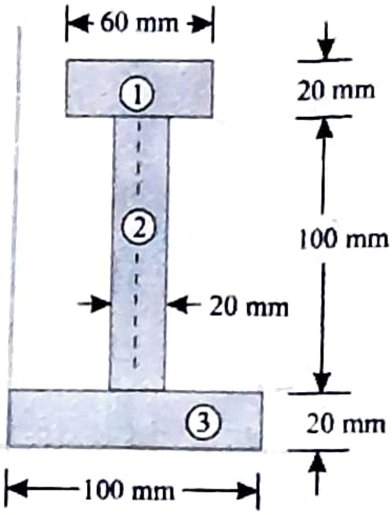

Figure 3

IV Determine the tension in the strings and accelerations of two blocks of mass 150 kg and 50 kg connected by a string and a frictionless and weightless pulley as shown in Fig 4 (10) L5 [CO3,4] 1,2

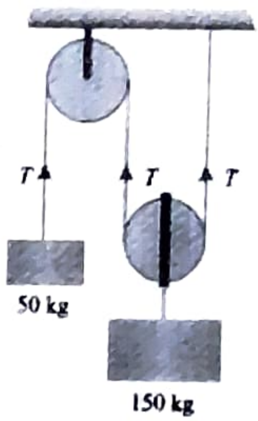

Figure 4

OR

V Three perfectly elastic balls A, B and C of masses 2 kg, 4 kg and 8 kg move in the same direction with velocities of 4 m/s, 1m/s and 0.75 m/s respectively. If the ball A impinges with the ball B, which in turn, impinges with the ball C, calculate the final velocities after impact (10) L5 [CO4] 1,2